Practice Problem Sheet-3

Course Code: PHY 2105/PHY 105

Course Title: Physics   Content: Coulomb's Law

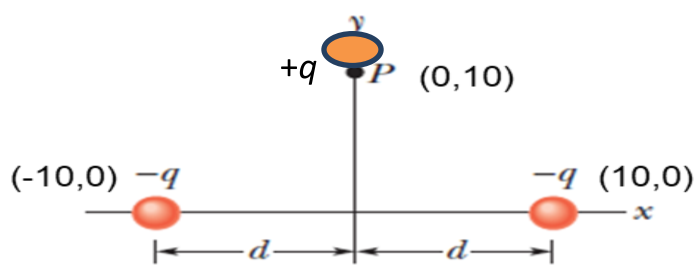
1. What is the electric force direction (draw the direction) at point q_1 if a distance r between point charge q_1 = 26.0 μC and point charge q_2 = -47.0 μC must be sustained for the electrostatic force between them to have a magnitude of 5.70 N? Find out the distance and express the equation along with direction of the force due by q_2. Express Coulomb force in vector form if the position of charges along X axis is $q_1(3,0)$ and the charge $q_2(7,0)$? Which type of force it is attractive or repulsive? What is the net electric force if a positive test charge +2 μC is introduced at the mid position between the two charges? What is the angle between the forces? Also, draw the direction of net electrostatic force.

2. A force $\vec{F}_1 = 3\hat{i} - 4\hat{j}$ is inclined at 60° to the horizontal. (i) If the horizontal component of force is 40N, calculate the vertical component. (ii) If another force $\vec{F}_2 = -2\hat{i} + 3\hat{k}$ is acting with a new angle which is also inclined with horizontal axis, then find out the angle between $\vec{F}_1$ and $\vec{F}_2$.

3. The x component of a force vector $\vec{A}$ is -25.0 m and the y component is +40.0 m. (i) What is the magnitude of the force vector? (ii) What is the angle between the direction of the force vector $\vec{A}$ and the positive direction of x?

4. Determine analytically the magnitude and direction of the resultant of the following four forces acting at a point of center of a coordinate system using superposition of forces principle. (i) 10 kN pull N 30° of E; (ii) 20 kN push S 45° of W; (iii) 5 kN push N 60° of W; and (iv) 15 kN push S 60° of E.

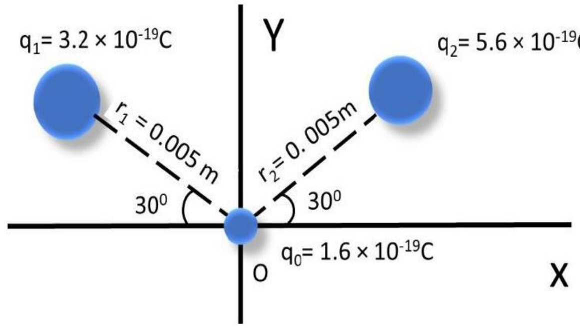
5. Two equal charges of 10×10^{-5} C are shown in fig below; each produces an electrostatic force at point P on y axis. (i) What is the magnitude of the force at P? (ii) What is the direction of Coulomb's force?



6. Two small charged spheres repel each other with a force = 3×10^{-3} N. The charge on one sphere is twice that on the other. When one of the charges is moved 10 cm away from the other, the force = 5×10^{-4} N. Calculate the charges and the initial distance between them.

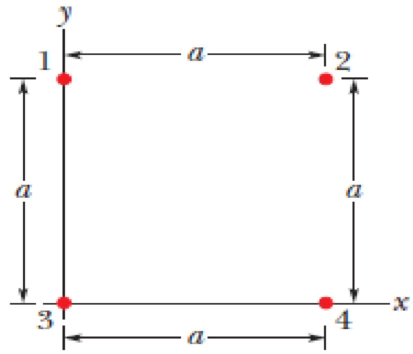
7. Four charges +2q, +4q, +2q and –2q are placed at the corners of a square. (i) Draw the arrangement of the charges. (ii) Calculate the magnitude and direction of electrostatic forces on a charge -1q at the intersection of the diagonals of the square of side 10 cm if q = 3×10^{-9}C.

8. From the figure, (i) Calculate the magnitude of net force on test charge q_0. (ii) Calculate the direction of net

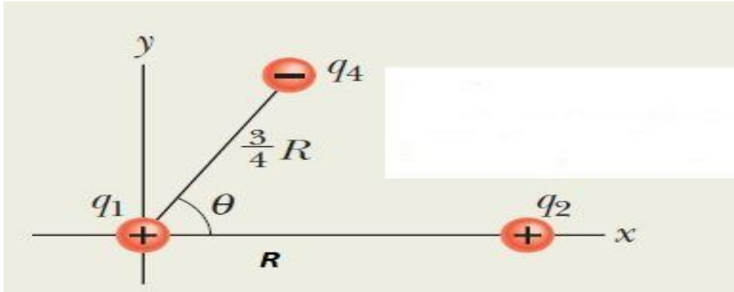


force on test charge q_0.

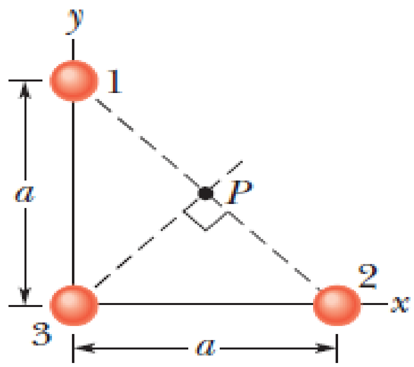
9. Four charges are arranged in a square with sides of length 2.5 cm. The two charges in the top right and bottom left corners are $+3.0\text{x}10^{-6}$ C. The charges in the other two corners are $-3.0\text{x}10^{-6}$ C. What is the net force and direction exerted on the charge in the top right corner by the other three charges?
10. What is (i) the force between two 4 gm pennies 2 m apart if we remove all the electrons from the $^{23}_{11}Na$ atoms? (ii) What is their acceleration as they separate? [Use quantization of charge principle]
11. How many charges will be available in 1 kg of electrons? [Use quantization of charge principle]
12. In the following radioactive decay reaction, $^{230}_{90}Th + \, ^0n \rightarrow \, ^{226}_{x}Ra + \alpha +$ energy , write down the value of x and α, and also show that charge is conserved. [Use conservation of charge principle]
13. Three charges lie on the x axis: q_1=+25 nC at the origin, q_2= -12 nC at x =2m, q_3=+18 nC at x=3 m. What is the net force on q_1? and What is the direction? [Use superposition of force principle]
14. Consider the two protons are separated at a distance 5 nm from each other. Compare the electrostatic force and gravitational force between them. The gravitational constant is G = 6.6743×10^{-11} m^3 $kg^{-1}s^{-2}$ and the mass of each proton is 1.67262192×10^{-27} kg. [Use Coulomb force Law and Newton's Gravitational Law]
15. Charges of magnitude 100 µC each are located in vacuum at the corners A, B and C of an equilateral triangle measuring 4 meters on each side. If the charge at A and C are positive and the charge B negative, what is the magnitude and direction of the total force on the charge at B?
16. Charges of magnitude 20 nC each are located in vacuum at the corners A, B and C of an right angle triangle measuring 5 m in height and 3 m base BC. If the charge at A and C are positive and the charge B is negative, what is the magnitude and direction of the total force on the charge at A?
17. What is the magnitude of the electrostatic force between a singly charged sodium ion (Na+, of charge +e) and an adjacent singly charged chlorine ion (Cl-, of charge -e) in a salt crystal if their separation is 3.81×10^{-7} m.
18. Point charge of + 8.0 µC and - 5.0 µC are placed on an x axis, at x = 8.0 m and x= 14 m. What charge must be placed at x = 26.0 m so that any test charge (q, q>0) at x = 0 experience no electrostatic force?
19. How far apart must two protons be if the magnitude of the electrostatic force acting on either one due to the other is equal to the magnitude of gravity force on a proton at Earth's surface?
20. In Fig below, the particles have charges $q_1 = -q_2$=100 nC and $q_3 = -q_4$= 200 nC, and distance a=5.0 cm. What are the (i) x and (ii) y components of the net electrostatic force on particle 3?



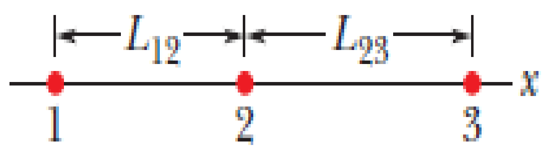
21. Two charges +5μC and +10μC are placed 20 cm apart. What is the electrostatic force at the midpoint between the two charges if a positive test charge of +5μC is introduced?
22. Two point charges +4q and +q are placed 30 cm apart. At what point on the line joining them the electric force is zero?
23. Two point charges q_A=3μC and q_B= −3μC are located 20 cm apart in vacuum.
    (i) What is the electric force at the midpoint O of the line AB joining the two charges?
    (ii) If a negative test charge of magnitude 1.5× 10^{-9} C is placed at this point, what is the force experienced by the test charge?
24. Two charges +5μC and -10μC are placed 10 cm apart. What is the electrostatic force at the midpoint between the two charges if a positive charge of +1μC is introduced? What will be happen if a negative charge of same magnitude is introduced?
25. Positive charges of 2 μC and 8 μC are placed 15cm apart. At what distance from the smaller charge will be the Coulomb force due to them be zero? Suppose the test charge inserted in between them is $+q_o$.
26. Two forces act at a point in directions inclined to each other at 120°. If the bigger force is 5N and their resultant is at right angles to the smaller force, find the resultant and the smaller force.
27. Of the charge *Q* initially on a tiny sphere, a portion *q* is to be transferred to a second, nearby sphere. Both spheres can be treated as particles and are fixed with a certain separation. For what value of *q*/*Q* will be the electrostatic force between the two spheres be maximized?
28. Three charged particles q_1, q_2, q_4 are arranged as in figure below. Calculate the net effective force on 1 due to particle 2 and 4. Given, $q_1 = 1.60\text{x}10^{-19}$ C , $q_2 = 3.2\text{x}10^{-19}$ C , $q_4 = -3.20\text{x}10^{-19}$ C , R=0.02m and $\theta = 60^0$.



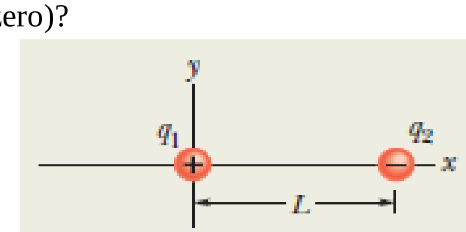
29. The three particles are fixed in place and have charges q_1= q_2= +*e* and q_3=+2*e*. Distance *a*= 6 m. What are the (i) magnitude and (ii) direction of the net electric force on a charge q_0= +*e* at point *P* due to the particles?



30. Two balls are each given a static electric charge of one ten-thousandth of a coulomb. (i) Calculate the force between the charges when they are separated by one-tenth of a meter. (ii) Compare the force with the weight of an average 70 kg person.
31. In the figure, three charged particles lie on an X axis. Particles 1 and 2 are fixed in place. Particle 3 is free to move, but the net electrostatic force on it from particles 1 and 2 happens to be zero. If $L_{23}=L_{12}$, what is the ratio q_1/q_2?



32. A positive charge of 6×10-6 C is 0.04 m away from the second positive charge of 4×10-6 C in +X direction. Calculate the force between the charges. Express the electrostatic force in vector. The force is attractive or repulsive? Draw the net force direction on first positive charge.
33. In an early model of the hydrogen atom (the Bohr model), the electron orbits the proton in uniformly circular motion. The radius of the circle is restricted (quantized) to certain values given by $r = n^2 a_0$, for n=1,2,3,….. where $a_0 = 52.29\ pm$. What is the speed of the electron if it orbits in (i) the smallest allowed orbit? and (ii) the second smallest orbit? (iii) If the electron moves to larger orbits, does its speed increase, decrease, or stay the same?
34. Three charges lie on the x axis: q_1= +40 µC at the origin, q_2= -28 µC at x =5 m, q_3= +30 µC at x=8 m. Calculate the (i) magnitude of net force on q_3 and (ii) direction of the net force on q_3 .
35. How many electrons would have to be removed from a coin to leave it with a charge of +1.0x10^{-7} C?
36. Figure shows two particles fixed in place: a particle of charge q_1=8q at the origin and a particle of charge q_2= -2q at x=L. At what point (other than infinitely far away) can a proton be placed so that it is in *equilibrium* (the net force on it is zero)?



37. The nucleus in an iron atom has a radius of about 4.0 x10^{-15} m and contains 26 protons. (i) What is the magnitude of the repulsive electrostatic force between two of the protons that are separated by 4.0 x10^{-15} m? (ii) What is the magnitude of the gravitational force between those same two protons?
38. How many charges are available in 1 kg of NaCl?